

Tinjau

Verified background checks for AI agents: facts about agents and their reviewers, proven from Ethereum into Creditcoin

Before one agent pays another, check what can actually be proven about it

BUIDL CTC 2026 Fall · Track: AI · Creditcoin CC3 Testnet + Attestcoin

Why a background check, and why on Creditcoin

- Creditcoin began as a credit history for people banks cannot see (Credal: 5M+ loans, 337k users)
- AI agents are the next "unbanked": **19,000+** on ERC-8004, hired and paid by other agents, with no history a contract can check
- Tinjau is the bureau in Creditcoin's sense, not the lender or the judge: it records **proven facts**, names the **known gaps**, and the party taking the risk sets the price
- The hiring escrow is the pricing step: its premium is the agent's **cost of credit**

The problem (Ethereum mainnet, measured 29 Aug 2026)

- **346 of 367** rated agents have exactly one reviewer; one wallet wrote **225** reviews for 195 agents
- **16 of 105** reviewers own agents and wrote **59%** of all feedback
- 60 days: **83%** of 14,771 registrations from owners holding ≥ 10 agents
- **0 of 12** sampled "paid on Ethereum" claims exist on Ethereum
- The spec's answer: "aggregation will happen off-chain", so trust an aggregator

Facts, not scores

- GroundedFacts admits data **only** through Attestcoin proofs of Ethereum txs from the official registries
- Reviews, reviewer seniority, holes in review indices, negatives, clones by owner / registrant / URI / minting tx, reviewer-owns-agents, **attestors registered at admission**
- No weights. The consumer passes thresholds; anyone recomputes the same numbers from the same proofs

Attestcoin depth (all measured on-chain)

Surface	Use
BlockProver 0x0FD2 verify	every proof, in-contract; 33 source txs admitted
calculateTxIndex	dedup across scouts (2nd cycle: 0 gas); exact transfer ordering
AttestorStash 0x0FD4	attestor count at admission; hirer can refuse thin quorums
ChainInfo 0x0FD3	attested tip in-contract; hirer can refuse stale facts
Depth and reach	history back to Mar 2022 (786 roots, 507k gas); the same proof verifies on CC3 mainnet (127,746 gas)

Money follows facts

- `AgentHireEscrow` : $\text{premium} = \text{base} + (\text{max} - \text{base}) \cdot \text{risk}$, paid **to the agent's owner**; `Gated` if a reviewer's record has holes
- `CoverageBounty` : pays only for proofs that **change the decision in that call**; hurting evidence pays the same as helping evidence
- No reserve, no oracle, no admin

Who fills the bureau in

- Anyone can bring proofs; nobody can invent one: the precompile checks every proof, logs count only from the official registries
- Tinjau's own scout runs unattended every 3 hours; other scouts are welcome and deduplicated
- A dishonest scout can only **withhold**, and the registry's review counter exposes the hole as a known gap
- Contributions change **completeness**, never **correctness**

The agent: GroundedScout (4 logged decisions)

- **R1** open bounties first, then requested or most-reviewed agents
- **R2** evidence both ways: complete records of senior reviewers; clones, conflicted reviewers, negatives
- **R3** skip what is admitted; prove only when the bounty covers gas
- **R4** hire on its own thresholds, else fund a bounty
- Live: claimed bounty #0 with the decisive batch, hired 2 agents at 1%, refused a gated one
- **Track fit:** verified cross-chain data in, autonomous on-chain transactions out, **no centralized oracle operator anywhere in the path**

Demo (mainnet data, Creditcoin txs, 13 Sep 2026)

- Agent **22771**: 3 reviewers active 97 days to 4 years before their first review, premium **1%**, hired
- Agent **21548**: bounty claimed by the scout, then hired at **1%**
- Agent **50283**: one reviewer who owns agents, review #97 proven and #1–96 not, 9 clone siblings, quote **20%**, `hire` reverts `Gated(1)`
- `tinjau_verify` replays all 33 admitted txs from chain data: identical

AI where rules cannot keep up

- Review documents claim payments in marketplace-specific formats; Gemini reads them and proposes (network, txHash) pairs, falling to the next model on a quota error
- Deterministic checks decide: the hash must be in the document, the chain must be Attestcoin-readable, prover and precompile confirm inclusion
- Live on agent 50283: **6 claimed payment transactions, 0 found on the chain they name**
- A report, never a fact on-chain; MCP tools let any agent ask the bureau before hiring

Limits we state up front

- Ethereum mainnet only (Sepolia dropped on purpose: free-to-mint entries); Base, where most ERC-8004 activity is, is out of reach today
- Facts are **admitted evidence with known gaps**, never a complete history
- Attestor count = attestors **registered at admission**, not the signers of one proof;
coveredThrough = newest proven fact, not "all blocks checked"
- Clone density is a lower bound; aged wallets can be bought
- Attestcoin moves trust to bonded attestors; it does not remove it (mainnet min bond is 0)
- Evidence Exchange: decision-change payout and source-order resolution are live; batch claim, adjudication receipt and same-tx re-pricing are not

What's next

- **Finish Evidence Exchange:** batch claim, an adjudication receipt (decision before and after, predicate flipped), hire re-pricing in the same transaction
- **A consumer contract on Creditcoin** beyond our own escrow: any marketplace, lender or agent already reads `facts()` through `IAgentFacts`
- **More source chains** as Attestcoin reaches them; the registry table is a constructor argument, so no rewrite
- **Who pays:** the hirer, as a premium on the job, and it goes to the agent's owner. A clean record is worth keeping

Team & links

- Dien: solo builder. Veritas Protocol (two-track winner, 9th Uniswap Hook Incubator); Vista (2nd place, Monad Blitz)
- Repo: <https://github.com/scientivan/Tinjau> · Live: <https://tinjau.xyz>
- Contracts (CC3 testnet): [0x67394eC13E911ab0D3A26132BECa404F26e17a98](#) ,
[0xF801a8a01E018f3Bf9a648F4C095a53979282EEA](#) ,
[0xa27f14CD50BF334E7Fb09601cEf203745aADF569](#)
- Video:
https://drive.google.com/drive/folders/13l7mJelLLJGnGH0y8Uy9PkM0aBbbMRKNq?usp=drive_link
- Integration summary: [ATTESTCOIN_INTEGRATION.md](#) · Dossier: [docs/evaluation-dossier.md](#)